N83 XOR cache: launch recovery and controls

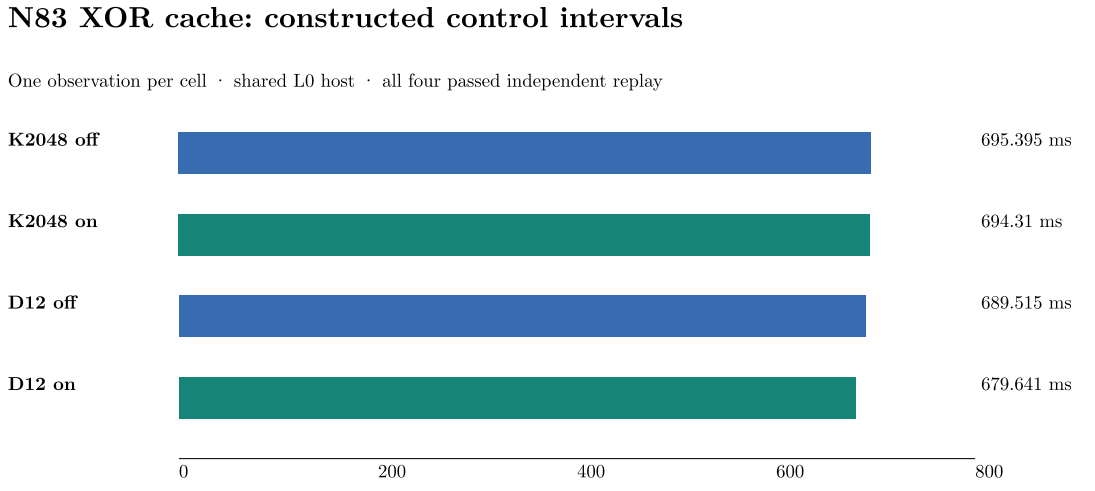
2026-10-20 · xor-cache-v2 · constructed correctness gate

All four actual controls—K2048/D12 with cache off/on—passed, each with a separate native general-field replay and rank one. Off/on returned identical complete SAT-model and relation-row digests within each base. The corrected card started at 01:16:34 UTC and entered public search after all controls passed.

Retained launch history

Start UTC	Observed state
01:08:07	Pinned image absent; exit 1 before worker execution. No natural query.
01:10:50	K2048 off control/replay passed; a global shell label caused a doubled replay path. Exit 2 before natural queries.
01:16:34	Exact image restored; driver function isolated in a subshell; four controls and four replays passed. Public card running at this boundary.

Both failed launch trees remain archived. The actual shell card passed two native fixture tests covering all 24 calls in order, a retained wall cap, remaining-cell execution and refusal to reuse outputs. Fixtures test driver sequencing; actual group correctness is checked independently.



One observation per cell, all with zero decisions/conflicts and pinned x blocks/selectors/phases. Cache-on skipped 3,735 satisfied-row visits in each control. This records activation; natural-query effectiveness remains pending. Different bases use different constructed targets. These shared-host L0 observations admit no speedup or base ranking.

Identity, limits and validation

Primary curve `icv1-f2m83-tm6151469093347-debefd74`, a=0, subgroup 241785163923079621668568 9, cofactor 4, modulus z^83+z^45+z^2+z+1. K2048: 2,048 columns / 339,968 points; D12: 2,001 / 332,166. Native worker revision `1de47a9d4186bff08abc84c224a111183e820106`, identical binary across settings. Independent general-field replay checks subgroup/phase/sign/coefficients and rejects tampering.

SSH Spot host, Intel Xeon 6975P-C, CPU 31, one CPU quota, hard 4 GiB, zero swap, no worker network and read-only inputs/root. Worker intervals include import, model/root/search, group checking and rank. Construction, Docker launch, independent replay and failed-launch overhead are outside the worker interval and separately retained. The elapsed field ends before final receipt serialization; container times preserve the process boundary. Other host workloads and evidence copying make these L0 observations.

Linux release library: 4,403 passed, zero failed, 118 pre-existing ignored. Worker 3, receipt audit 2, replayer 2 and native card fixture 2 release tests passed. Cache invalidation and 400 mixed CNF/XOR truth-table cases passed. The cache stays off by default.

Sources and open gates

Study `research/n83_prebuild_cost_m23_20261019/`: `XOR_CACHE_PROTOCOL.md`, host card, both continuation notes, `XOR_CACHE_VALIDATION.md`, and this directory's `RUN_REPORT.md`. Control archive `xor-cache-v2/controls/controls.tar.xz`: SHA-256 `fcaa18926f019f0fcc9fdd739827e58beaa8ce675bac355aa276656468c24f84`, 104 checked files. Failure archives: 19 and 39 checked files under `xor-cache-v1/`. Plot reads the four unmodified worker receipts in `data/`.

The sixteen-search public comparison uses targets 0 and 1, both bases, A1/B1/B2/A2, 10,000 conflicts and 1,790/1,800 s caps. Returned relations require independent replay. Natural rank, complete cold/reused IC cost, matched rho and best-base selection remain open. Candidate and best-runtime fields stay null; canonical leaderboard/boundary rows are unchanged.